

Bayesian Learning - Computer Lab 1

Liuxi Mei

Xiaochen Liu

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1. Daniel Bernolli

Let $y_1, \dots, y_n \mid \theta \sim \text{Bern}(\theta)$, and assume that you have obtained a sample with $f = 35$ failures in $n = 78$ trials. Assume a $\text{Beta}(\alpha_0, \beta_0)$ prior for θ and let $\alpha_0 = \beta_0 = 7$.

(a)

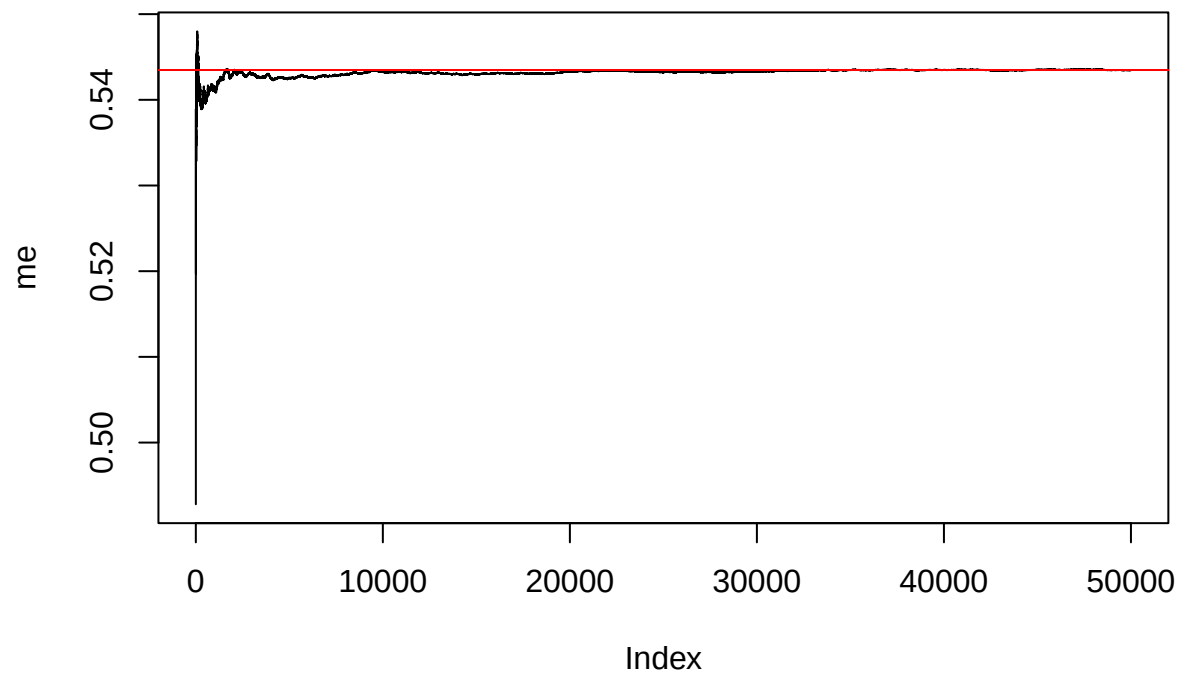
Draw 10000 random values ($n\text{Draws} = 10000$) from the posterior $\theta \mid y \sim \text{Beta}(\alpha_0 + s, \beta_0 + f)$, where $y = (y_1, \dots, y_n)$, and verify graphically that the posterior mean $\mathbb{E}[\theta \mid y]$ and standard deviation $\text{SD}[\theta \mid y]$ converges to the true values as the number of random draws grows large.

Give model is a bernoulli trial and prior has a $\text{Beta}(\alpha_0, \beta_0)$ distribution, posterior $\text{Beta}(\alpha_0 + s, \alpha_0 + f)$, where $s=(n-f)=53$ and $f=35$. Thus: posterior $\text{Beta}(\alpha_0 + s, \alpha_0 + f)$ posterior $\text{Beta}(50, 42)$

```
nr <- 50000
set.seed(123456)
res <- rbeta(nr, 50, 42)
me <- numeric(nr)
std <- numeric(nr)

for (i in 1: nr) {
  me[i] <- mean(res[1:i])
  std[i] <- sqrt(var (res [1:i]))
}

plot(me, type = 'l')
abline(h=50/(50+42), ,col = 'red')
```



```
plot(std, type = 'l')  
abline(h=sqrt(50*42/(50+42)^2/(42+50+1)),col = 'red')
```

